Robotic Interference with Crowd Evacuation
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Clustering
The clustering approach used in this project is K-Means clustering with a k value of 2. The arms regard the 2 nearest clusters and choose the largest one to focus on and repel. This generally works, though it struggles sometimes if multiple individual spheres are approaching the same exit from different directions. It is, however, a good compromise between simplicity of implementation and computation and effectiveness.

Prediction Model
The prediction model used for anticipating the motion of the escaping spheres is a simple linear extrapolation model. The velocity of the spheres is calculated and a new position is estimated based on their current velocity and position. Since the spheres largely move at a constant speed, particularly near exits, this is a fairly good predictor of their motion. The prediction is constantly being recalculated in order to ensure the arms maintain awareness of the spheres' positions.

Crowd Interaction
Due to the K-Means algorithm used, the robot arms tend to target clusters of spheres that are heading towards the exit and split them down the middle. In doing this, the arms repel the spheres back away from the exit, while the spheres still try to make their way towards the exit. This can lead to success, and in some circumstances the arm can keep a given sphere away from an exit indefinitely. However, as elaborated on in the Observations sections, the arms can become their undoings as they break up clusters and subsequently cannot cope with the full spread of spheres from different directions.

Observations
As I refined and observed the simulation, I found that the arms tended to break up larger clusters and split them down the middle. This lead to an interesting phenomenon where many of the spheres would get around the arm by splitting up into smaller and smaller groups until he arm could no longer keep them away because it was spending too much time oscillating between individual spheres. There would also be times when the arm would get too focused on a single sphere while letting some others through as it took a little bit of time to move back to the other side of the exit. The combination of these factors leads to a high eventual success rate for escaping spheres reaching the exit.